LF2 Worked Examples: Equations of Lines

Recording companion for the LF2 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

Do I know a point and the slope? If not, what will get me there? Every equation of a line comes from a point and a slope. Point-slope form writes them straight in: $y - y_1 = m(x - x_1)$. From two points, find the slope first. From a graph, find two grid corners on the line. From standard form, solve for y: the number in front of x is the slope, and the constant is the y-intercept.

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Example 1: Standard Form 1

$$3x+4y=12$$

Question

What are the slope and the y-intercept of this line?

A. slope −3/4, y-intercept (0, 3)
B. slope 3/4, y-intercept (0, 3)
C. slope −4/3, y-intercept (0, 3)
D. slope 3/4, y-intercept (0, 12)

Answer and reasoning

Solve for y: $4y=-3x+12$, so $y=-(3/4)x+3$. The slope is the number in front of x, $-3/4$, and the y-intercept is $(0,3)$. A slope of $3/4$ loses the minus sign from moving $3x$ across; $-4/3$ divides the wrong way, B over A; and slope $3/4$ with $(0,12)$ makes two slips at once: it loses the minus and forgets to divide the 12 by 4. **Slope $-3/4$, y-intercept $(0,3)$.**

Example 2: Standard Form 2

$$2x-5y=10$$

Question

What are the slope and the y-intercept of this line?

A. slope −2/5, y-intercept (0, 2)
B. slope 2/5, y-intercept (0, 2)
C. slope 2/5, y-intercept (0, −2)
D. slope 5/2, y-intercept (0, 2)

Answer and reasoning

Move $2x$: $-5y=-2x+10$. Now divide every term by -5: $y=(2/5)x-2$. Slope $2/5$, y-intercept $(0,-2)$. With a negative B, the signs are where the mistakes happen: slope $-2/5$ with $(0,2)$ ignores the sign of $-5y$ and divides every term by $+5$; slope $2/5$ with $(0,2)$ gets the slope right but divides the 10 by $+5$; and slope $5/2$ with $(0,2)$ divides the wrong way, B over A, and also loses the sign of the intercept. **Slope $2/5$, y-intercept $(0,-2)$.**

Example 3: Standard Form 3

$$-6x-4y=10$$

Question

What are the slope and the y-intercept of this line?

A. slope 3/2, y-intercept (0, 5/2)
B. slope −3/2, y-intercept (0, −5/2)
C. slope −2/3, y-intercept (0, −5/2)
D. slope 3/2, y-intercept (0, −5/2)

Answer and reasoning

Move $-6x$: $-4y=6x+10$. Divide every term by -4: $y=-(6/4)x-10/4=-(3/2)x-5/2$. Both reduce, and the y-intercept does not have to be a whole number: it is $(0,-5/2)$, halfway between -2 and -3. Slope $3/2$ with $(0,5/2)$ divides by $+4$ instead of -4; slope $-2/3$ divides 4 by 6 instead of 6 by 4; and slope $3/2$ with $(0,-5/2)$ divides the 10 by -4 but forgets the sign when dividing $6x$. **Slope $-3/2$, y-intercept $(0,-5/2)$.**

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Example 4: Standard Form 4

$$-7x+y=0$$

Question

What are the slope and the y-intercept of this line?

A. slope −7, y-intercept (0, 7)
B. slope 7, y-intercept (0, 7)
C. slope 7, y-intercept (0, 0)
D. slope −7, y-intercept (0, 0)

Answer and reasoning

Add $7x$ to both sides: $y=7x$. The slope is 7, and since there is no constant, $b=0$: the line goes through the origin, so the y-intercept is $(0,0)$. CheckIt's C can be 0, so this happens. Slope -7 with $(0,0)$ forgets that moving $-7x$ across changes its sign; slope 7 with $(0,7)$ uses the slope as the intercept, but with no constant the intercept is 0; and slope -7 with $(0,7)$ makes both slips. Every non-vertical line has a y-intercept, even when it is 0. **Slope 7, y-intercept $(0,0)$.**

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Example 5: Point and Slope

$$m=-3, \quad (4,-2)$$

Question

Which is an equation of the line with this slope through this point?

- **A.** $y=-3x-14$
- **B.** $y=-3x+10$
- **C.** $y=-3x-2$
- **D.** $y=-3x-10$

Answer and reasoning

Point-slope form: $y-(-2)=-3(x-4)$, so $y+2=-3x+12$ and $y=-3x+10$. Check: $-3(4)+10=-2$. -14 comes from $-2=-12+b$ worked as $-2=12+b$; -2 puts the point in backwards, $4=-3(-2)+b$; and -10 flips both signs inside, $y-2=-3(x+4)$. ***y=−3x+10*, or *y+2=−3(x−4)*.**

Example 6: Two Points 1

$$(2,-3) \text{ and } (-4,4)$$

Question

Which is an equation of the line through these points?

A. $y=-(7/6)x-2/3$
B. $y=-(6/7)x-9/7$
C. $y=(7/6)x-16/3$
D. $y=-(7/6)x-16/3$

Answer and reasoning

Slope first: $m=(4-(-3))/(-4-2)=7/(-6)=-7/6$. Then point-slope with $(2,-3)$: $y+3=-(7/6)(x-2)$, so $y=-(7/6)x+7/3-3=-(7/6)x-2/3$. Check the other point: $-(7/6)(-4)-2/3=14/3-2/3=4$. $-(6/7)x-9/7$ puts run over rise, then finds b from $(2,-3)$ with that slope; $(7/6)x-16/3$ subtracts the y-values in one order and the x-values in the other, $(4-(-3))/(2-(-4))$; and $-(7/6)x-16/3$ has the right slope but solves $-3=-7/3+b$ as $b=-3-7/3$. ***y=−(7/6)x−2/3*, or *y+3=−(7/6)(x−2)*.**

Example 7: Point-Slope Form

$$m=5, \quad (-3,7)$$

Question

Which is the point-slope form of this line?

- **A.** $y+7=5(x-3)$
- **B.** $y-7=5(x-3)$
- **C.** $y-3=5(x+7)$
- **D.** $y-7=5(x+3)$

Answer and reasoning

Point-slope form subtracts the point's coordinates: $y-y_1=m(x-x_1)$. With $x_1=-3$, $x-(-3)=x+3$; with $y_1=7$, $y-7$. So $y-7=5(x+3)$. $y+7=5(x-3)$ flips both signs; $y-7=5(x-3)$ ignores that -3 is negative; and $y-3=5(x+7)$ swaps the coordinates. CheckIt accepts this form or slope-intercept, $y=5x+22$. ***y−7=5(x+3).***

Example 8: Two Points 2

$$(-8,-3) \text{ and } (7,7)$$

Question

Which is an equation of the line through these points?

A. $y=(3/2)x+9$
B. $y=(2/3)x+7/3$
C. $y=(2/3)x-7/3$
D. $y=(2/3)x+16/3$

Answer and reasoning

$m=(7-(-3))/(7-(-8))=10/15=2/3$. With $(-8,-3)$: $y+3=(2/3)(x+8)$, so $y=(2/3)x+16/3-3=(2/3)x+7/3$. Check the other point: $(2/3)(7)+7/3=21/3=7$. The fraction intercept is normal for two random points; point-slope form, $y+3=(2/3)(x+8)$, is just as correct. $(3/2)x+9$ flips the slope; $-7/3$ writes $y-3=(2/3)(x-8)$; and $+16/3$ forgets to subtract the 3. ***y=(2/3)x+7/3*.**

Example 9: Graph 1

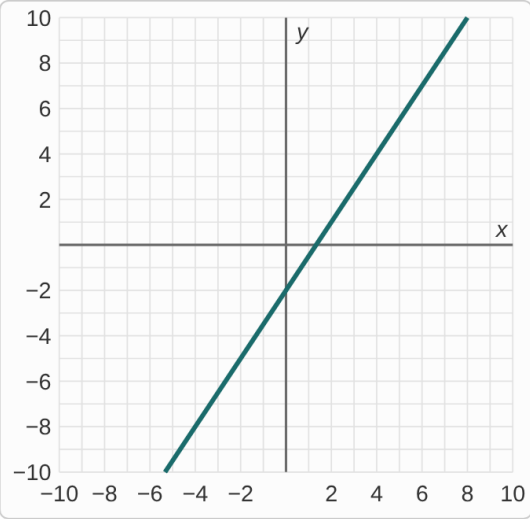


Figure: A straight line on a grid from −10 to 10 in both directions, with grid lines every 1 unit and no points marked. It rises from left to right, passing through (0, −2), (2, 1) and (4, 4).

Question

Which is an equation of this line?

- **A.** $y=(2/3)x-2$
- **B.** $y=-(3/2)x-2$
- **C.** $y=(3/2)x-2$
- **D.** $y=(2/3)x+1$

Answer and reasoning

Grid corners at $(0,-2)$ and $(2,1)$: over 2, up 3, so $m=3/2$. The line crosses the y-axis at -2: $y=(3/2)x-2$. Check: $(3/2)(4)-2=4$. $2/3$ is run over rise; $-3/2$ has the wrong sign for a rising line; and $(2/3)x+1$ makes two slips: run over rise, and the height of the point $(2,1)$ used as if it were the y-intercept. ***y=(3/2)x−2*.**

Example 10: Graph 2

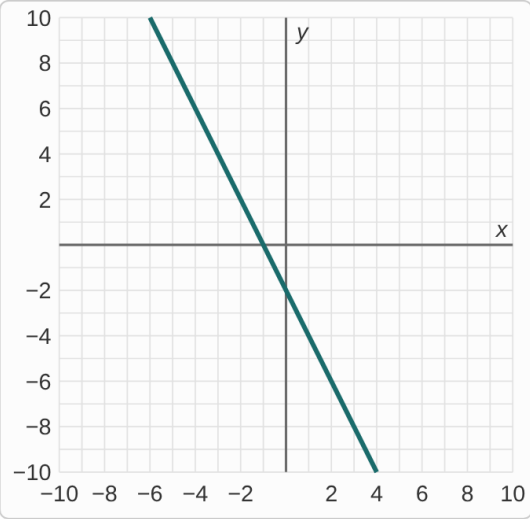


Figure: A straight line on a grid from −10 to 10 in both directions, with grid lines every 1 unit and no points marked. It falls steeply from left to right, passing through (−4, 6), (−3, 4) and (0, −2).

Question

Which is an equation of this line?

- **A.** $y=-2x-2$
- **B.** $y=2x-2$
- **C.** $y=-(1/2)x-2$
- **D.** $y=2x+4$

Answer and reasoning

From $(-3,4)$ to $(0,-2)$: over 3, down 6, so $m=(-6)/3=-2$. It crosses the y-axis at -2: $y=-2x-2$. Check: $-2(-3)-2=4$. $2x-2$ has the wrong sign for a falling line; $-1/2$ is run over rise; and $2x+4$ makes two slips: the wrong sign, and the height of $(-3,4)$ used as the y-intercept. ***y=−2x−2*.**

Example 11: Graph 3

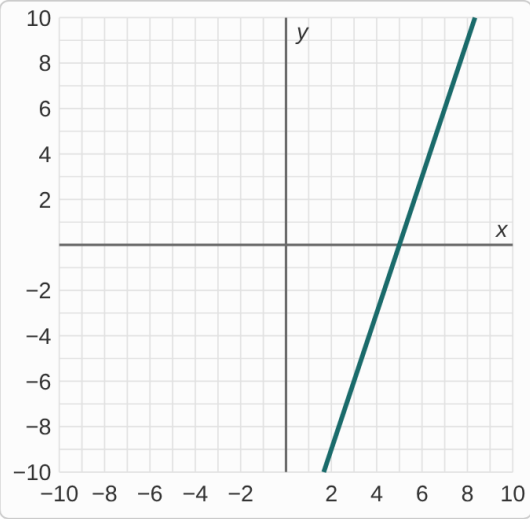


Figure: A straight line on a grid from −10 to 10 in both directions, with grid lines every 1 unit and no points marked. It rises steeply from left to right, passing through (4, −3), (5, 0) and (6, 3). It leaves the bottom of the grid before it reaches the y-axis.

Question

Which is an equation of this line?

- **A.** $y=3(x-5)$
- **B.** $y=3x+5$
- **C.** $y=3(x+5)$
- **D.** $y=(1/3)(x-5)$

Answer and reasoning

The y-intercept is off the grid, so use a grid corner the line does cross: $(5,0)$ on the x-axis. From $(5,0)$ to $(6,3)$: over 1, up 3, so $m=3$. Point-slope with $(5,0)$: $y-0=3(x-5)$, or $y=3(x-5)$. In slope-intercept form that is $y=3x-15$, which is why the y-intercept is off the grid. Check: $3(4-5)=-3$. $3x+5$ uses the x-intercept's 5 as if it were the y-intercept; $3(x+5)$ flips the sign inside, which is the line through $(-5,0)$; and $1/3$ is run over rise. **$y=3(x-5)$, or $y=3x-15$.**

Example 12: Graph 4

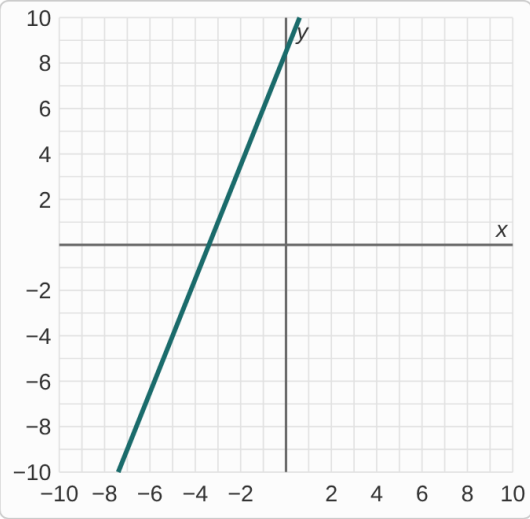


Figure: A straight line on a grid from −10 to 10 in both directions, with grid lines every 1 unit and no points marked. It rises steeply from left to right, passing through (−5, −4), (−3, 1) and (−1, 6), and crosses the y-axis between 8 and 9.

Question

Which is an equation of this line?

- **A.** $y-4=(5/2)(x-5)$
- **B.** $y+4=(2/5)(x+5)$
- **C.** $y-4=(2/5)(x-5)$
- **D.** $y+4=(5/2)(x+5)$

Answer and reasoning

This line crosses the *y*-axis between grid lines, so read a grid corner instead. From $(-5,-4)$ to $(-3,1)$: over 2, up 5, so $m=5/2$. Point-slope with $(-5,-4)$: $y+4=(5/2)(x+5)$. In slope-intercept form that is $y=(5/2)x+17/2$, and CheckIt accepts either. $y-4=(5/2)(x-5)$ flips both signs; $2/5$ is run over rise; and $y-4=(2/5)(x-5)$ makes both slips. ***y+4=(5/2)(x+5)*.**

Example 13: Slip 1

$$6x+2y=8$$
$$slope=6/2=3$$

Question

A student found the slope this way. What is the mistake, if any?

A. There is no mistake.
B. Solve for y: y = −3x + 4, so the slope is −3.
C. Slope is B over A, so it should be 2/6, or 1/3.
D. The slope is 8/2 = 4, the constant divided by B.

Answer and reasoning

Moving $6x$ to the other side changes its sign: $2y=-6x+8$, so $y=-3x+4$ and the slope is -3, which is $-A/B$. The student skipped the minus. $2/6$ divides the wrong way; and $8/2=4$ is the y-intercept's height, not the slope. **The slope is -3.**

Example 14: Slip 2

$$5x-2y=8$$
$$y\text{-intercept}=(0,8)$$

Question

A student found the y-intercept this way. What is the mistake, if any?

A. There is no mistake.
B. It should be (8, 0), since the 8 is on the right.
C. Divide by −2 too: the intercept is (0, −4).
D. Divide by 5 instead: the intercept is (0, 8/5).

Answer and reasoning

The 8 is not the intercept until y is alone: $-2y=-5x+8$, so $y=(5/2)x-4$. Setting $x=0$ in the original gives the same thing: $-2y=8$, $y=-4$. So the y-intercept is $(0,-4)$. $(8,0)$ puts the number in the x place, which is a different point; and $8/5$ divides by the x-coefficient, which gives the x-intercept's value, not the y-intercept's. **The y-intercept is $(0,-4)$.**

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Example 15: Slip 3

$$(1,5),\ (3,9)$$
$$m=(9-5)/(3-1)=2$$
$$5=2(1)+b,\ b=3$$
$$y=2x+3$$

Question

A student wrote this work. What is the mistake, if any?

A. There is no mistake.
B. The slope should be 1/2, run over rise.
C. b should come from the second point, so b = 9.
D. The equation should be y − 5 = 2(x + 1).

Answer and reasoning

The slope is $4/2=2$, and $5=2+b$ gives $b=3$. Check the other point: $2(3)+3=9$. The student is right. Slope is rise over run, so $1/2$ is upside down; either point gives the same b, and $9=2(3)+b$ also gives 3, not 9; and point-slope form with $(1,5)$ is $y-5=2(x-1)$, with a minus. **There is no mistake.**

Example 16: Slip 4

$$m=4,\ (-2,3)$$
$$y-3=4(x-2)$$

Question

A student wrote this equation. What is the mistake, if any?

A. There is no mistake.
B. It should be y + 3 = 4(x − 2), since the 3 is positive.
C. Swap the numbers: y − 2 = 4(x − 3).
D. x − (−2) is x + 2: y − 3 = 4(x + 2).

Answer and reasoning

Point-slope form subtracts the coordinates: $x-x_1=x-(-2)=x+2$. So $y-3=4(x+2)$, which is $y=4x+11$. The student's line goes through $(2,3)$ instead. $y+3$ would be for a point with y-coordinate -3; and swapping the coordinates puts the point at $(3,2)$. ***$y-3=4(x+2)$*.**

Example 17: Slip 5

$$m=-2,\ (3,1)$$
$$1=-2(3)+b$$
$$b=-5$$
$$y=-2x-5$$

Question

A student wrote this work. What is the mistake, if any?

A. There is no mistake.
B. 1 = −6 + b gives b = 7, so y = −2x + 7.
C. The point goes in backwards: 3 = −2(1) + b, so b = 5.
D. The slope should be 2, so y = 2x − 5.

Answer and reasoning

$1=-6+b$. Adding 6 to both sides gives $b=7$, not -5: the student subtracted. So $y=-2x+7$. Check: $-2(3)+7=1$. The point goes in as $x=3$, $y=1$, not the other way around; and the slope is given as -2, so it stays -2. ***y=−2x+7.***